Intervention-Aware Execution Adaptation around Frozen Manipulation Policies

Evolutionary selection, feedback timing and a direction-preserving recovery diagnostic

Sachin Mohan
University of the Witwatersrand
Johannesburg, South Africa
2699183@students.wits.ac.za

ABSTRACT

Can a compact execution controller improve safe completion around a frozen vision-language-action policy under a small interaction budget? We study joint-target restraint and policy-query timing in ManiGuard Clutter, retaining the native task and safety monitors. A 24-episode evolutionary search selects a four-gene controller. Its eight-block dial pilot was adverse: 4/8 safe completions and 2/8 violations at full strength, against identity's 5/8 and 0/8. A subsequent 120-episode comparison, 40 episodes per dial setting, reverses that sign. Safe completion rises from 20/40 at identity to 27/40 at half and 32/40 at full strength (Fisher exact $p = .009$, full versus identity), and violations fall from 8/40 to 4/40 and 2/40 ($p = .087$). A separate sixteen-episode feedback pilot finds no early activation of its correction trigger. We then implement a uniform, direction-preserving cap and evaluate it on 18,657 saved commands from 24 historical episodes. At 0.04 rad, componentwise clipping distorts joint-error direction by more than five degrees on 9/55 modified commands; vector scaling does so on 0/55. A frozen median correction threshold admits three early-query opportunities in 482 diagnostic prefixes, with sparse calibration support. The larger comparison supports an execution benefit for safe completion in the evaluated tasks; the violation reduction is suggestive rather than established, and neither result separates evolutionary search from matched random search or ordinary learning. Supporting forecast and warning-prediction audits explain why model-assisted search remains untested.

CCS CONCEPTS

• **Computing methodologies**;

KEYWORDS

evolutionary robotics, frozen policies, execution adaptation, safe manipulation

1 INTRODUCTION

A manipulation policy can complete a task while tipping a protected object or violating another physical requirement. Conversely, a controller can avoid a violation by doing little and leaving the task unfinished. ManiGuard makes this distinction explicit through independent task checks and runtime monitors over physics-grounded predicates [4]. This makes safe completion, violations and unfinished episodes necessary companion outcomes.

Our question is whether a small controller around a frozen policy can improve this trade-off with substantially less adaptation machinery than training the underlying policy. The controller changes arm-target magnitude and the frequency of observation feedback. Evolution searches its categorical parameters using complete simulator episodes. An external dial interpolates between the original policy and the selected controller. A useful result would require improved measured safe completion, a credible physical-violation comparison, and an advantage that survives matched random search and ordinary learning.

The completed evidence meets the first of these requirements and partly the second. The evolutionary finalist's eight-block dial pilot is adverse, but a subsequent 120-episode comparison of the same three dial settings raises safe completion from 20/40 under the frozen policy alone to 32/40 at full strength, with violations falling from 8/40 to 2/40. The feedback pilot does not activate its adaptive timing mechanism. Separately, the original componentwise cap motivates a repair with a precise software property: preserve the proposed joint-error direction when limiting target displacement. The new saved-trace experiment verifies that property and measures how rarely a correction-based refresh could have acted on the available records.

The paper contributes an implemented execution interface, a 120-episode comparison showing a safe-completion gain over the frozen policy, audited adverse and null pilot results, and a reproducible diagnostic linking a concrete geometry repair to its remaining physical test. It does not establish optimizer superiority, a calibrated safety dial, cross-task transfer or a safety guarantee. This internal draft consolidates completed work and explicitly identifies the unexecuted comparisons.

2 CONTEXT AND RESEARCH QUESTION

ManiGuard supplies specification-grounded manipulation tasks and separates success from safety [4]. We use its frozen Clutter policy and pinned openpi implementation [5]; we do not train a new VLA. Our adapter has no online access to protected-object poses, native grasp labels or monitor predicates.

Dynamics models can reduce the physical evaluations needed for skill search [2]. Frozen visual representations also support learned dynamics and planning [3, 6], while model-based policy optimization motivates testing where predictions remain useful [1]. These precedents make a new optimizer name or the use of visual features insufficient as a novelty claim. Our proposed contribution concerns execution intervention and the allocation of limited physical evaluations. The current study first asks whether the execution mechanism works, and separately whether the predictive prerequisites are adequate.

We distinguish three questions: does the execution map improve outcomes over identity; does adaptive refresh contribute beyond a fixed schedule; and does evolutionary selection improve over other adaptation methods at matched cost? The 120-episode dial comparison addresses the first. The feedback pilot addresses the second only in a small descriptive sample. The third remains open because matched random-search and ordinary-learning arms have not run.

3 PROBLEM AND EVALUATION BOUNDARY

At policy query time τ, a frozen policy produces a sequence $P_\tau = \pi(o_\tau; \xi)$ from two RGB views, native robot input and its task prompt. Between queries, the controller consumes cached proposals $p_t \in \mathbb{R}^8$. Their first seven coordinates are arm joint targets and the eighth is the gripper proposal after native binarization. Measured arm joints are $q_t \in \mathbb{R}^7$. The adapter produces a_t and native action-space clipping produces the final environment input u_t. We log p_t, a_t and u_t separately. The simulator's actuator processing is a further boundary; a bound on $u_t - q_t$ is not a bound on joint speed, contact force, end-effector motion or object tilt.

Let V be the episode's native raw `ltl_violated` flag. Requested completion C_{req} combines native success with the pinned Clutter completion checklist. Safe completion and the training utility are

$$S_{\text{req}} = C_{\text{req}} \wedge \neg V, \qquad U = \mathbf{1}(S_{\text{req}}) - 2\,\mathbf{1}(V). \tag{1}$$

Thus a violating episode receives −2, an unfinished nonviolating episode receives zero, and a safe completion receives one. This is a declared penalized utility. It is not penalty-free optimization. Reporting retains the four joint categories: safe completion, unsafe completion, safe unfinished and unsafe unfinished.

The monitor labels consequences and never changes commands (Figure 1). Throughout this draft, "execution controller" refers to our map and query scheduler. There is no learned online safety filter, world-model veto or ROSARL integration in the executed system.

4 METHODOLOGY

4.1 Original categorical execution controller

Part 1 searches $\theta = (c_f, c_g, \alpha, h)$ over

$$\begin{aligned} c_f &\in \{\varnothing, .12, .08, .04\} \text{ rad}, \quad & c_g &\in \{\varnothing, .08, .04, .02\} \text{ rad}, \\ \alpha &\in \{1, .75, .5\}, & h &\in \{4, 8, 16\}. \end{aligned} \tag{2}$$

The domain contains 144 genotypes. At the first action $r_t = q_t$; subsequently $r_t = u^{\text{arm}}_{t-1}$. The anchor computes

$$z_t = r_t + \alpha(p^{\text{arm}}_t - r_t), \qquad a^{\text{arm}}_t = q_t + \text{clip}(z_t - q_t, -c, c), \tag{3}$$

or $a^{\text{arm}}_t = z_t$ when the selected cap is disabled. The gripper proposal is preserved. A fixed causal heuristic chooses c_g when both absolute finger positions are in $[.003, .035]$ m, maximum finger change is below .002 m, and the current proposal closes the gripper. Otherwise it chooses c_f. This heuristic is robot evidence rather than a held-object detector.

An episode-constant dial $d \in [0, 1]$ applies

$$\tilde{a}^{\text{arm}}_t = p^{\text{arm}}_t + d\,(a^{\text{arm}}_t - p^{\text{arm}}_t), \qquad h(d) = \lfloor 8 + d(h - 8) + .5 \rfloor. \tag{4}$$

At zero, an explicit bypass returns the original proposal bitwise. At one, the anchor is returned directly. The experiment searches at $d = 1$ and evaluates the frozen finalist at $d \in \{0, .5, 1\}$. Native gripper handling, goals, monitor rules and early success termination remain unchanged.

Algorithm 1 Executed Part 1 evolutionary search

Require: Frozen policy, domain Θ, sealed cases and seeds
$P_0 \leftarrow$ identity plus three distinct uniform points
for $g = 0, 1, 2$ **do**
  for candidate θ_i in P_g **do**
    Run two fresh full simulator episodes at dial one
    Audit both records; set $F_i \leftarrow (U_{i,1} + U_{i,2})/2$
  end for
  Rank by decreasing F_i, then stable birth order
  if $g < 2$ **then**
    Retain top two parents; copy best into P_{g+1}
    Add three distinct categorical mutants of those parents
  end if
end for
Freeze best candidate in the final evaluated population
Evaluate its three dial settings on eight matched blocks
Ensure: Full lineage, all outcomes and all interaction costs

4.2 Evolution with fresh elite evaluation

Algorithm 1 gives the executed Part 1 search. PCG64 seed 20261018 initializes identity and three distinct uniform random genotypes. Each population has four candidates, each evaluated on two training cases. After stable ranking by mean utility, the best two candidates become parents. The next population physically re-evaluates the best parent and evaluates three new children; historical elite fitness is never copied into a new generation.

Mutation selects one of the two parents uniformly and designates one coordinate for mutation. Each other coordinate is visited with probability .5. A visited coordinate is uniformly resampled with probability .2; otherwise its index moves by one of $\{-2, -1, 1, 2\}$ and is clipped to the categorical range. Rejection removes unchanged offspring and duplicates within the new population. After at most 100 draws, a declared uniform fallback selects an unused eligible point. These are index operations on the declared domain, not Gaussian mutations of physical units. Invalid infrastructure records have no fitness and cannot enter ranking.

Only the final evaluated population supplies the finalist. Search uses tasks 0034/0002, with seed $1000 + 10g + i$ for task index i. The 24 pilot comparison episodes use tasks 0029/0032, four repetitions each, at all three dial settings. Their seeds are $2000 + 10j + r$. A separate schedule RNG interleaves conditions within blocks. These scenes are conservatively considered exposed; the pilot supplies fresh repetitions rather than new-task transfer evidence.

A later dial comparison evaluates 40 episodes at each of the same three settings, 120 in total. **[TODO: state the evaluated controller (Part 1 finalist [3,3,0,2] or the repaired map), tasks, seeds, condition schedule, whether n=40 was fixed before outcomes, and whether the 120 include the 24 pilot episodes]**

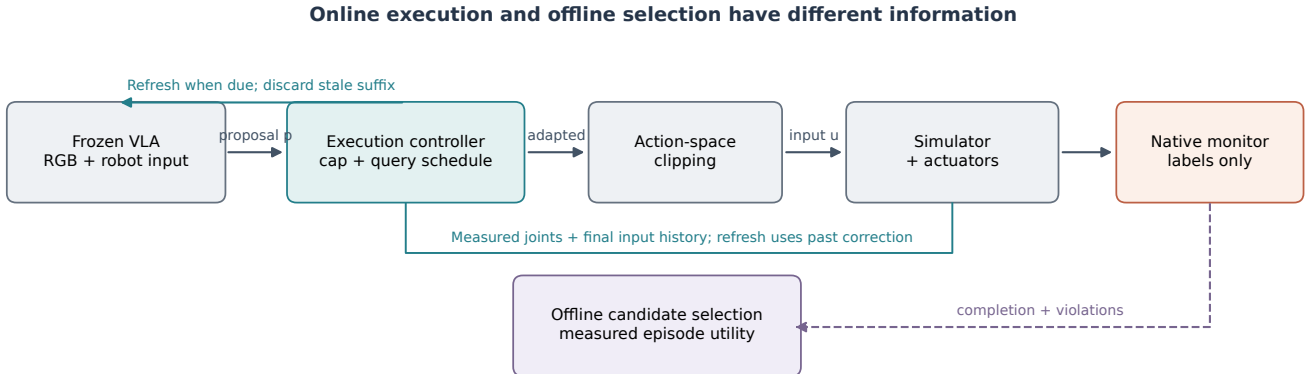


Figure 1: Information flow. The execution controller receives policy proposals, measured robot state and past final environment inputs. Native monitor outcomes enter offline evaluation and candidate selection. The monitor does not veto or replace an action. Final environment input precedes lower-level actuator processing.

Algorithm 2 Causal correction-aware execution interface

Reset accumulated correction and cached policy prefix
while episode has not terminated **do**
  if no prefix, prefix exhausted, or refresh is due **then**
    Query frozen policy at current observation
    Discard old suffix; reset $D \leftarrow 0$, $k \leftarrow 0$
  end if
  Read next proposal p_t and measured joints q_t
  $a_t \leftarrow$ declared cap map; preserve gripper
  $u_t \leftarrow$ native action-space clipping of a_t
  Apply u_t; log proposal, adaptation and final input
  Update D using Eq. (5); $k \leftarrow k + 1$
  Refresh due if $k = 8$ or $(k \geq m \wedge D \geq \eta)$
end while

4.3 Correction-aware feedback scheduling

Part 2 removes smoothing and the long sixteen-action interval, retains the finalist's .04/.02 rad caps, and compares identity/8 (A), cap/8 (B), cap/4 (C), and adaptive cap/4–8 (D). The adaptive scheduler accumulates normalized departure between the proposal and the final environment input:

$$D_k = \sum_{t=\tau}^{\tau+k-1} \frac{\|\mathcal{T}(p_t^{\text{arm}} - u_t^{\text{arm}})\|_2}{\sqrt{7}\,(.04\text{ rad})}, \tag{5}$$

where $\mathcal{T}$ zeros per-coordinate differences no larger than 10^{-6} rad. The sum uses already applied inputs, includes downstream clipping and resets at each policy query. Finger history persists. With minimum interval four, D queries when $D_k \geq .5$ or at maximum age eight, subject to native termination taking precedence. A zero threshold is a fixed-frequency reference; a disabled threshold is fixed-eight execution. Correction measures command departure, not physical risk.

The sixteen-episode schedule uses two repetitions of tasks 0039/0010 and rotates ABCD, BCDA, CDAB and DABC across four blocks. It is a separate feedback pilot, not a new evolutionary search. The same passive observation capture runs in all conditions. No trigger threshold is retuned during that physical comparison.

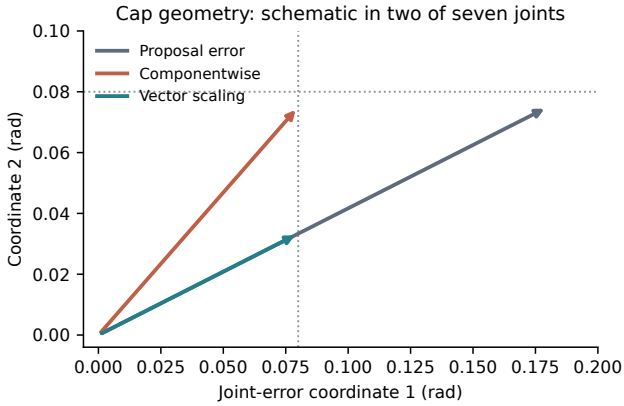


Figure 2: A schematic of the two cap maps. Both limit the largest joint-error coordinate; uniform scaling retains the original error direction. This is a two-coordinate illustration, not a simulated or measured robot trajectory.

4.4 Uniform direction-preserving recovery map

The new recovery implementation fixes smoothing to one and removes grasp conditioning. For joint-error vector $e_t = p_t^{\text{arm}} - q_t$, define

$$s_t = \begin{cases} 1, & \|e_t\|_\infty = 0, \\ \min(1, c/\|e_t\|_\infty), & \text{otherwise}, \end{cases} \qquad a_t^{\text{arm}} = q_t + s_t e_t. \tag{6}$$

For $c > 0$, scalar multiplication preserves the joint-error direction and bounds every coordinate's target error by c before numerical rounding. Componentwise clipping can rotate that direction when coordinates saturate differently (Figure 2). This property concerns joint-space targets. It does not preserve Cartesian motion or certify

Algorithm 3 Saved-trace trigger-opportunity diagnostic

Require: Verified records, fixed cap, calibration/diagnostic tasks
for each complete factual eight-action calibration prefix **do**
  Map saved p_t, q_t with Eq. (6); apply saved bounds
  Compute cumulative correction using these mapped inputs
  Collect positive D_k for $k \in \{4, 5, 6, 7\}$
end for
Freeze the declared three quantile thresholds
for each complete factual eight-action diagnostic prefix **do**
  Find first $k \in \{m, \ldots, 7\}$ with $D_k \geq \eta$
  Count one opportunity if such k exists
  Make no claim about proposals after that hypothetical query
end for
Ensure: Opportunity counts, support, task denominators and zero new physics

collision avoidance or uprightness. Downstream clipping and actuator processing can alter the physical interpretation. The disabled branch returns the original proposal exactly.

The saved-trace protocol fixes disabled/.12/.08/.04 rad caps and evaluates both maps on all 24 verified full Part 2/3 recordings. Every proposal is transformed at its factual measured joints, preserving its recorded dtype and gripper. We replay the recorded action-space bounds and measure direction before and after that boundary. Engineering episodes are excluded. None of these alternate inputs is executed.

For the timing diagnostic, the cap is fixed at .04 rad before scoring. Calibration uses the eight B/D episodes on tasks 0010/0039. Only complete factual eight-action prefixes are eligible. We collect strictly positive accumulated correction at ages four through seven and derive thresholds from their .50/.75/.90 quantiles. The median is the prospectively selected reference. A diagnostic panel uses only the four fixed-eight B recordings on tasks 0046/0006, excluding four-action schedules and incomplete final prefixes. Both minimum intervals four and six are reported. All four tasks were exposed previously, so this panel is development evidence, not a fresh recovery audit.

The first would-trigger age is an open-loop diagnostic within a factual prefix. After that point a real policy query would produce new proposals and a different trajectory. We therefore infer neither counterfactual completion nor total new query cost from this replay. Threshold support and per-task activation are reported alongside pooled counts.

4.5 Supporting prediction audits

Part 3 separately tests whether a compact predictive system is useful enough to help allocate evolutionary evaluations. A frozen DINOv2-small encoder supplies 2×2 spatially pooled features from two cameras, 3,072 coordinates in total [3]. Fitting-only standardization and PCA retain 32 visual coordinates; arm joints and fingers remain separate. Five episode-bootstrap 128/128-unit residual MLPs predict standardized visual/robot changes from current and previous state plus previous/current executed inputs. Training is fixed at 30 epochs. Causal histories shift during rollouts of at most four actions.

Table 1: Cumulative physical costs, counted once. Unknown initialization physics and one pre-application failure prevent treating these as exact totals. Dial-comparison controls and queries await its run ledger; if its 120 episodes include the 24 Part 1 pilot episodes, those are already counted in Part 1.

Stage	Attempts	Controls	VLA queries
Part 1, all work	52	35,158	3,401
Part 2, all work	20	11,286	1,796
Part 3, recordings	8	7,467	1,385
Part 4, offline	0	0	0
New recovery replay	0	0	0
Known total, excl. dial comp.	80	53,911	6,582
Dial comparison (§6.2)	120	**TODO**	**TODO**

Ridge readouts predict upright margin and negative target-to-goal distance. The audit separates readout error on actual future states from error after predicted dynamics. References include persistence, ridge and shuffled-action tests. Disagreement-only trust and a correction-conditional factual-error rule are compared before any physical use. Object-state targets are evaluation labels and never become online controller inputs. A distance readout is distinct from native held completion; a tilt readout is distinct from spill or drop safety.

Part 4 instead fits a direct warning predictor for the first native uprightness event within eight actions, with a warning deadline four actions before onset. Its causal inputs include current visual features, robot/proposal history, four past executed inputs, controller metadata and query age. A five-member 64-unit MLP ensemble uses 500 updates per member; logistic and visual-only references are retained. Probability and alarm rules calibrate on a separate task at natural prevalence with episode weights. This is a warning-classification audit, not a repaired latent world model or an executed counterexample search.

5 EXPERIMENTAL PROTOCOL AND ACCOUNTING

All physical episodes use the pinned ManiGuard Clutter setup, two 256×256 camera inputs, native gripper binarization, assisted grasping, raw joint control, 20 Hz action/render and 120 Hz physics. The frozen checkpoint is step 7100 at revision `f572e33bdb49`; the full revision and file hashes remain in the controller export. The native horizon is 1,100 policy actions, with native early termination. No episode is shortened to obtain a favourable completion count.

Historical physical execution used separate uv-managed Python 3.10 simulation and Python 3.11 policy environments on one RTX 4090. The saved-trace recovery experiment instead runs on a different RTX 4090 preparation machine without an installed ManiGuard stack, as CPU replay of private revision-pinned recordings. That replay creates no new physics, policy queries, encoder calls, fitted predictors or physical outcomes. The 120-episode dial comparison is physical execution. **[TODO: machine, environments and checkpoint revision check for the dial comparison]**

The original model partitions use sixteen Part 2 episodes for fitting, four task 0046 episodes for calibration and four task 0006

episodes for audit. Once exposed, these tasks cannot become fresh recovery evaluation. The main-study test partition remains excluded. All 24 historical acquisitions are charged once, including discarded candidates and recording costs; reuse adds analysis cost rather than new physical experience. Table 1 retains the cumulative lower bounds.

Matching task and sampling seed does not guarantee identical observations. All sixteen Part 1 initial robot-state comparisons agree numerically, but both cameras differ in every matched pair, and the first policy commands already differ before adaptation. Part 2 likewise has equal initial robot states but different camera images and first chunks. These checks prevent interpreting a single paired video as an identified causal counterfactual.

6 RESULTS

6.1 Evolution's training winner is adverse in the eight-block pilot

The 24 search episodes evaluate twelve candidate measurements, including elite re-evaluations. Five measurements attain maximum utility. The selected genome is [3, 3, 0, 2]: free cap .04 rad, grasp cap .02 rad, smoothing one and horizon sixteen. It safely completes its two final training cases. The copied elite's fresh mean score falls to .5. Two-case fitness is therefore visibly noisy, and the finalist's perfect training score is not an estimate of general performance.

In the 24 pilot comparison episodes, safe completion is 5/8 at identity, 3/8 at midpoint and 4/8 at full strength (Figure 3, centre). Violations are 0/8, 2/8 and 2/8. Full minus identity is a descriptive −12.5 percentage points with two additional violating episodes. Full strength completes six requested tasks, but two completions are unsafe. Its 312 queries are fewer than identity's 774; that resource saving accompanies worse measured safety and safe completion in this pilot. The altered query horizon and arm map are confounded here.

An exploratory failure audit identifies a grasp-heuristic gap: in one unsafe liquid transport episode, it misses all 167 held-and-closing actions because a finger lies outside the fixed band. Other safe episodes also contain misses, and both obstacle-uprightness failures occur while the heuristic recognizes the grasp. The gap is concrete but is not a sufficient causal explanation. A timeout likewise does not distinguish inactivity, poor continuation, a restrictive controller or insufficient horizon.

6.2 A 40-episode-per-setting comparison favours full strength

The larger comparison evaluates 40 episodes at each dial setting (Figure 3, left; Table 2). Safe completion is 20/40 (50.0%) at identity, 27/40 (67.5%) at half strength and 32/40 (80.0%) at full strength. Full minus identity is +30.0 percentage points (Newcombe 95% CI 9.1 to 47.6; Fisher exact $p = .009$). Violations are 8/40, 4/40 and 2/40; full minus identity is −15.0 points (95% CI −30.2 to −0.1; Fisher exact $p = .087$). Both outcomes order monotonically with dial strength (Cochran–Armitage trend $p = .005$ for safe completion and $p = .037$ for violations). Half strength does not differ from identity at conventional levels on either outcome ($p = .17$ and $p = .35$).

Table 2: Dial comparison, 40 episodes per setting. Wilson 95% intervals in percent, treating episodes as independent.

Setting	Safe completion	Violations	Completed
Identity	20/40 (35–65)	8/40 (10–35)	24/40
Half	27/40 (52–80)	4/40 (4–23)	29/40
Full	32/40 (65–90)	2/40 (1–17)	33/40

The gain is not bought by inactivity. Requested completion rises from 24/40 to 33/40 ($p = .047$), unfinished episodes fall from 16/40 to 7/40, and unsafe completions fall from 4 of 24 completions to 1 of 33.

This sign reverses the eight-block pilot. If the 120 episodes include the pilot's 24, the 32 additional episodes per setting would contain 15, 24 and 28 safe completions and 8, 2 and 0 violations; identity's violation count in particular would move from 0/8 to 8/32. **[TODO: state whether the pilot episodes are included and, if so, account for the identity shift; add per-task outcomes and query counts per setting]**

All tests treat episodes as independent; repeated scenes make them optimistic. If the evaluated controller is the Part 1 finalist, its sixteen-action horizon and arm map remain confounded, as in the pilot.

6.3 The original adaptive trigger never refreshes early

Identity, cap/8 and adaptive cap/4–8 each safely complete 3/4 episodes; cap/4 achieves 2/4. Each condition records one violation. All 337 observed inter-query intervals for adaptive D are eight actions. Its sole .5-threshold crossing occurs at action eight, when a query is already compulsory. There is no early crossing at ages four through seven. This is a nonactivation result, not a comparison of an active adaptive mechanism with a fixed scheduler.

Fixed-four uses 750 queries against fixed-eight's 336 without improving this small sample's safe completion. This does not show that frequent feedback causes harm. It does establish that assuming an efficiency benefit from more frequent queries would be unsupported by the measured pilot.

6.4 Recovery fixes command geometry, with sparse timing opportunity

The new saved-trace screen uses 18,657 historical commands across 24 episodes on four tasks. With a uniform .04 rad cap, both maps substantially change 55 commands. Componentwise clipping rotates joint-error direction by more than 5° on 9/55 changed commands, with maximum 10.32°. Vector scaling has 0/55 above 5° and maximum 0.00019°, consistent with float32 rounding (Figure 4). No saved action-space bound further changes the arm command in this panel. Lower-level actuator output limits are outside this replay.

Only 55/18,657 commands, about .30%, are substantially modified at .04 rad. At .08 and .12 rad, the corresponding counts are three and one. These rates describe proposals along the original trajectories. They do not establish how often a deployed repaired controller would intervene after its states diverge.

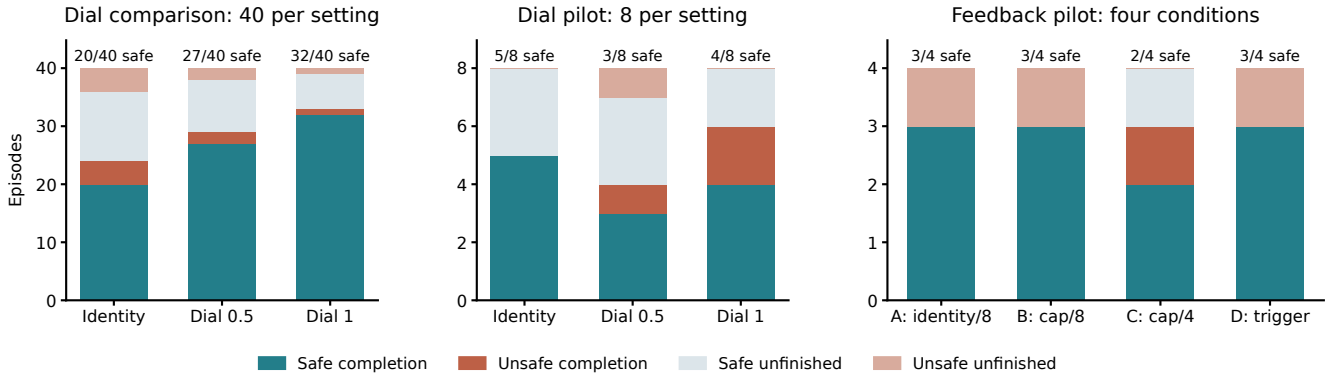


Figure 3: Every physical comparison outcome. Left: the 120-episode dial comparison, 40 episodes per setting. Centre: the Part 1 pilot, eight repetitions per setting across two scenes. Right: Part 2, four episodes per condition across two scenes. All four completion/violation categories remain visible. The comparison favours full strength on safe completion; none of these panels establishes an optimizer advantage.

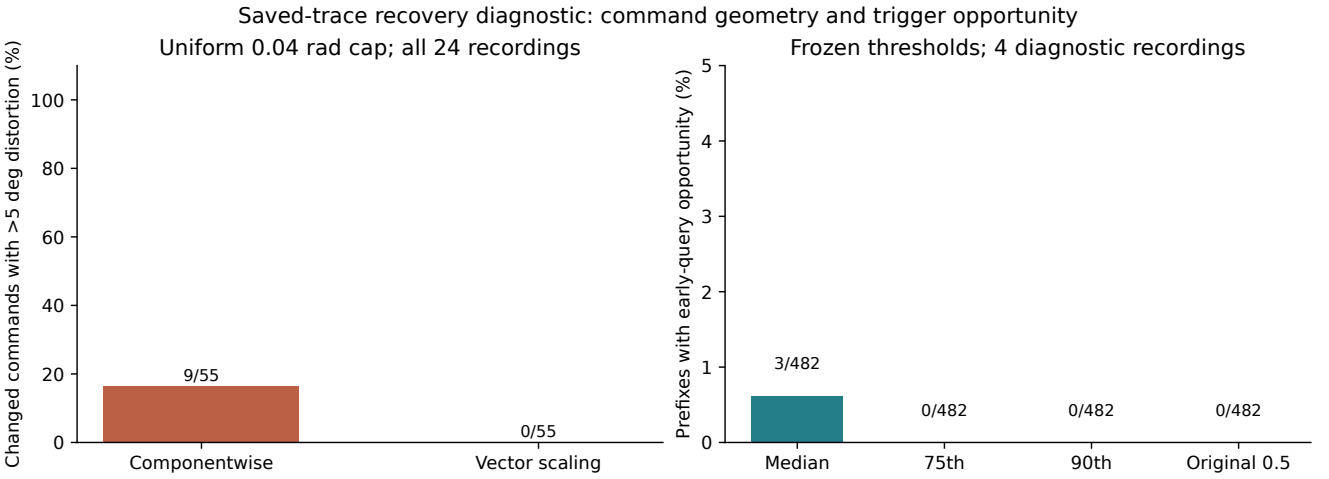


Figure 4: New saved-trace diagnostic. Left: direction distortion conditional on a substantially modified command at the .04 rad cap. Right: first early-query opportunities in 482 complete factual eight-action diagnostic prefixes, using thresholds frozen from calibration traces. Opportunity counts are neither actual queries nor physical completion measurements. All tasks are previously exposed.

Calibration has 669 complete eight-action prefixes but only five strictly positive eligible accumulated-correction values. These correlated values yield thresholds .089753, .536969 and .947503. The median threshold gives 3/482 early opportunities on the diagnostic panel, present in two of four episodes and only task 0046. The upper thresholds and original .5 reference give 0/482. This indicates a possible timing opportunity with weak cross-task support. It does not justify a claim of robust activation, saved query cost or improved physical safety.

The analysis completes in approximately eight CPU seconds after data restoration and implementation. It performs zero new simulator controls, physics steps, policy queries, encoder calls or trained-model fits. A repeated analysis after lint/source-preservation cleanup produces identical geometry and timing tables; both run receipts and source snapshots remain available. Twenty-one targeted checks cover identity, finite inputs, dtype, cap geometry, downstream clipping, causal correction, query reset and the age-eight boundary.

6.5 Prediction prerequisites remain unsupported

Part 3's four-action latent prediction error is 45.1% worse than persistence. The actual-state uprightness readout already has 16.17°

mean absolute error, before model rollout; near-threshold predicted-state error is 18.56°. All correction calibration lies in one supported bin, so the correction-dependent trust mask equals the disagreement-only mask. The planned 64-episode model-assisted optimizer comparison is unstarted. Failed readout, failed dynamics and indistinguishable selection rules are separate findings.

Part 4's frozen direct warning predictor alarms on 2,045/2,229 negative development windows, or 91.75%. Development contains one first-event episode and no confident miss. The event support and operating point fail the premise for its conditional physical acquisition design. No evolutionary counterexample-acquisition outcome, diversity effect or predictor-repair benefit is claimed.

7 DISCUSSION AND THE NEXT CONTROLLED COMPARISONS

The 120-episode comparison supports a safe-completion benefit from executing the frozen policy through the controller at full strength, within the evaluated tasks. The violation reduction points the same way but is not established: its exact test gives $p = .087$, and 2/40 violations retains an exact one-sided 95% upper bound of 14.9% even under independence. The result does not show that evolution found this controller more efficiently than alternatives. If the evaluated controller is the Part 1 finalist, it also does not separate the arm map from the longer query horizon. The first priority is therefore evidential rather than algorithmic: attach the run ledger, report outcomes per task, and repeat the frozen comparison on fresh tasks with caps, dial settings and sample size fixed in advance.

The saved-trace evidence separately supports a direction-preserving implementation and identifies the original trigger's lack of opportunity. The sparse .04 rad intervention rate means that a physically inconclusive repair screen is plausible. The proposed repair should therefore be judged by observable controller behaviour first, and by complete outcomes second, without changing caps or thresholds until a desired sign appears.

The recovery plan's smallest physical screen is eight full episodes: two preselected development tasks crossed with identity, componentwise uniform cap/8, vector cap/8, and vector cap with active refresh. All share the same cap and frozen policy. The cap and threshold must be fixed before those outcomes; the sparse threshold support above must remain disclosed. The screen is a mechanism test, not a powered safety or optimizer comparison. A later timing-efficiency claim also needs fixed-four/fixed-eight and a query-allowance control with actual query counts, all costed in advance.

Evolutionary value requires the next comparison to include matched random search and episodic ordinary learning over the same categorical controller and information. The recovery plan proposes a categorical REINFORCE-style update with a baseline, entropy regularization and fixed learning rate. These ordinary-learning updates have not been implemented or run in this check-in. Copied elites and forced identity anchors cannot be treated as uncorrected score-function samples. All methods must receive the same shared initial data, descendant evaluation allowance and confirmation cases, with finalists frozen before independent audit. One optimizer seed would still support only a pilot.

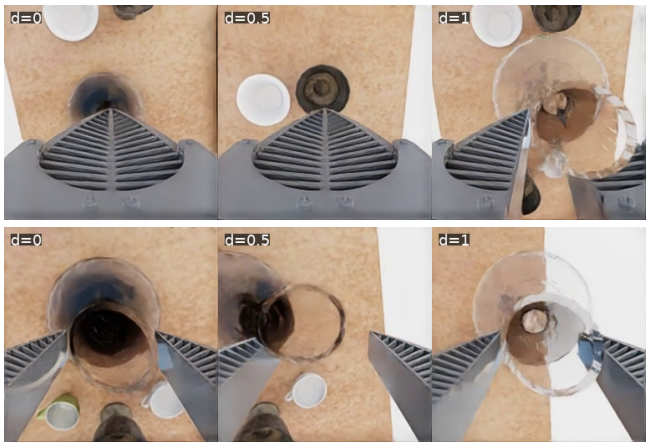


Figure 5: Historical native wrist frames from the eight-block pilot at the same 18-second timestamp, columns identity, midpoint and full dial. Top: full strength later completes safely while the other settings time out. Bottom: both adapted settings violate uprightness; full strength later completes unsafely. Selection is post hoc and illustrative. The complete eight-block gallery and newly saved local matched videos retain the positive and adverse context. No alternate trajectory is synthesized.

The principal inferential limitation remains the task denominator. Repeated frames, prefixes and episodes within a small number of scenes are correlated. We do not substitute them for independent task or optimizer-seed replication, and every reported test is optimistic for that reason. The pilot-to-comparison reversal itself shows how far eight-episode estimates can move. No safety noninferiority claim follows from an absence of a significant difference, and no transfer claim follows from repetitions within evaluated scenes.

8 REPRODUCIBILITY AND CURRENT STATUS

The repository retains the original frozen protocols, controller exports, public attempt ledgers, native outcome audits and complete historical video galleries. Private raw recordings and checkpoint lineage are revision-pinned and hash-verified. The new recovery protocol fixes input roles, cap domain, threshold derivation, prefix eligibility, a ten-minute analysis limit and zero new physical allocation. Its outputs include source snapshots, frozen numerical thresholds, per-episode geometry/timing tables and a public diagnostic receipt. Plotting and typesetting do not replay physics. **[TODO: attach run receipts, seeds, checkpoint revision and native monitor trace hashes for the 120-episode dial comparison; the Acknowledgments' trace statement now covers these episodes]**

The check-in deliverables are this paper draft, the 120-episode dial comparison, the completed offline diagnostic, the tested recovery controller and annotated historical videos. The original eleven-page cumulative evidence report remains available separately. Physical repair comparisons, new evolutionary runs, random-search/ordinary-learning controls, fresh-task replication of the dial

comparison, fresh-task recovery evaluation and replicated optimizer evidence remain future work.

9 CONCLUSION

A compact evolutionary controller is inexpensive to represent. Its eight-block dial pilot was adverse, but a 120-episode comparison of the same settings measures a 30-point safe-completion gain at full strength over the frozen policy alone, with fewer violations that do not yet reach significance. The original adaptive trigger never refreshes early, and the completed recovery diagnostic verifies a precise geometry repair while revealing sparse timing support. The next results must show that the dial gain holds on fresh tasks with a complete run ledger, separate evolutionary selection from matched random search and ordinary learning, and execute the geometry repair in a prospectively fixed physical comparison. Current evidence provides a concrete implementation, a measured execution benefit in the evaluated scenes, and a testable mechanism.

ACKNOWLEDGMENTS

This work is supervised by Geraud Nangue Tasse at the University of the Witwatersrand. AI coding assistance was used for implementation, analysis, prose and typesetting. All reported physical outcomes derive from recorded native simulator/monitor traces. The new replay diagnostic is explicitly separate from those physical measurements. Experimental pictures and videos use recorded native images; schematic figures are labelled as such. The archived studies and original recorded outcomes are unchanged.

REFERENCES

[1] Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. 2019. When to Trust Your Model: Model-Based Policy Optimization. https://doi.org/10.48550/arXiv.1906.08253 arXiv:1906.08253

[2] Bryan Lim, Luca Grillotti, Lorenzo Bernasconi, and Antoine Cully. 2021. Dynamics-Aware Quality-Diversity for Efficient Learning of Skill Repertoires. https://doi.org/10.48550/arXiv.2109.08522 arXiv:2109.08522

[3] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, et al. 2023. DINOv2: Learning Robust Visual Features without Supervision. https://doi.org/10.48550/arXiv.2304.07193 arXiv:2304.07193

[4] Yiyan Peng, Philip Wang, Simon Sinong Zhan, Yiqi Lyu, Zhenyang Ni, Jixin Yan, Fiorelli Wong, Ruochen Jiao, Hang Yin, Xinyu Cao, Huajie Shao, Manling Li, Ruohan Zhang, and Qi Zhu. 2026. MANIGUARD: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation. https://doi.org/10.48550/arXiv.2608.17386 arXiv:2608.17386 [cs.RO]

[5] Physical Intelligence. 2026. *openpi: Pinned policy implementation.* https://github.com/Physical-Intelligence/openpi/tree/215abfb217dbac7d5f1273282331b9b1866c0479 Revision 215abfb217dbac7d5f1273282331b9b1866c0479; accessed 18 October 2026.

[6] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. 2024. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. https://doi.org/10.48550/arXiv.2411.04983 arXiv:2411.04983